
Playing Devil’s Advocate: Off-the-Shelf Persona Vectors Rival Targeted Steering for Sycophancy

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Abstract

We study the effect of different persona on **sycophancy**: model’s agreement with users even when the user is incorrect. The standard mitigation, Contrastive Activation Addition (CAA), derives a steering direction from labelled pairs of sycophantic and honest responses. This study evaluates whether off-the-shelf persona steering vectors, originally developed for general role-playing and not trained on sycophancy data, can serve as an alternative. In two instruction-tuned models, steering toward personas characterised by doubt or scrutiny reduces sycophancy to approximately 68% and 98% of CAA’s effect, and, unlike CAA, maintains accuracy when the user is correct. The effect is also asymmetric: steering toward agreeable personas does not produce a mirror increase in sycophancy. Geometrically, the persona vector is largely independent of the direction of sycophancy in activation space. Collectively, these findings suggest that sycophancy is better understood as a persona-level property rather than a single steerable direction. We release our code here: <https://anonymous.4open.science/r/Sycophancy-Steering-9DF0/>.

1. Introduction

Sycophancy is the tendency of large language models to agree with users regardless of factual correctness. It is among the most persistent failure modes of RLHF-trained systems (Perez et al., 2022; Sharma et al., 2023). Even when a model has encoded the correct answer internally,

the reward model’s preference for agreement can override it (Wang et al., 2025a). How can we intervene on sycophancy without expensive retraining or curated behavioral datasets?

Contrastive Activation Addition (CAA) (Rimsky et al., 2024; Turner et al., 2023) extracts a steering vector from contrastive sycophantic/honest prompt pairs and adds a scaled version during inference. While effective, CAA requires hundreds of behavior-specific pairs and must be re-curated for each new target behavior. A natural alternative is to reuse *persona* representations already learned by instruction-tuned models (Lu et al., 2026).

We ask three questions. (i) Can off-the-shelf role vectors reduce forced-choice sycophancy comparably to CAA? (ii) Do critical/conformist family labels predict steering direction? (iii) Are effective role vectors geometrically distinct from the CAA direction? We evaluate across Gemma 2 27B (Gemma Team, 2024) and Qwen 3 32B (Qwen Team, 2025) on a counterbalanced PhilPapers benchmark with tune/test splitting and Holm correction. Overall, we make 3 novel contributions:

1. Critical-role vectors reach 68–98% of CAA’s Δlogit with strong cross-seed consistency, using *no* sycophancy labels.
2. Conformist roles produce weak, heterogeneous effects, partially falsifying bidirectional family-level prediction.
3. All role vectors are nearly orthogonal to CAA ($|\cos| < 0.17$), but the *sign* of the cosine flips between Gemma and Qwen — a cross-model geometric asymmetry we make explicit as a new caveat on mechanistic-independence claims.

2. Related Work

Perez et al. (2022) introduced model-written sycophancy evaluations exposing systematic agreement bias. Sharma et al. (2023) showed sycophancy is widespread and emerges during RLHF. Wang et al. (2025a) traced sycophancy to override mechanisms in RLHF-trained models: the correct answer is often encoded internally but suppressed by preference for agreement. More broadly, recent work shows that surface behavioral metrics can underestimate the extent of

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internal representational change during behavior modification (Chaudhary & Barez, 2025).

Activation addition (Turner et al., 2023) demonstrated that fixed residual-stream vectors can steer LLM behavior at inference time. CAA (Rimsky et al., 2024) formalizes this via mean-difference contrasts on labelled A/B pairs. Related work includes representation engineering (Zou et al., 2023), inference-time intervention (Li et al., 2023), conditional activation steering (Lee et al., 2025), and applications of steering to safety properties such as reasoning-trace leakage in chain-of-thought (Batra et al., 2025). Goral et al. (2025) study depth-wise steering across layers, and recent work provides a unifying causal-abstraction framework for these intervention techniques (Geiger et al., 2025).

Persona directions. Lu et al. (2026) introduced the Assistant Axis and released per-role steering vectors (Skeptic, Judge, Peacekeeper, . . .) derived from generation-plus-judge contrastive pipeline. Feng et al. (2026) compose persona directions via vector algebra for inference-time personality control. Pai et al. (2026) merge persona vectors. Wang et al. (2025b) link persona features to emergent misalignment. Relatedly, different post-training procedures leave distinct mechanistic footprints on the internal structures encoding model behavior (Nunez et al., 2026), suggesting that persona representations may be differentially shaped by the adaptation objective used. Such procedure-dependent behavioral shaping can also propagate subliminally through benign training data during distillation (König et al.).

Persona-sycophancy link. Shah et al. (2026) showed persona *agreeableness* correlates with sycophancy at r up to 0.87. Vennemeyer et al. (2025) argue sycophancy decomposes into causally separable components along distinct linear directions. Our work sits at this intersection: we test whether *off-the-shelf* role vectors — never trained on sycophancy labels — transfer to a held-out forced-choice benchmark, and we characterize the geometric relationship to a targeted CAA direction on two models. The broader question of when behaviors decompose into modular versus entangled internal structures has been studied through training-procedure interventions that aim to produce more modular circuit organization (Golechha et al., 2025).

3. Methodology

3.1. Models and target layers

We use Gemma 2 27B Instruct (Gemma Team, 2024) and Qwen 3 32B (Qwen Team, 2025) because they are instruction-tuned decoder-only models at comparable scale but with substantially different baseline sycophancy rates on our benchmark (59% vs. 84%), providing a natural robustness test. Steering is applied at layer 22 of 46 (Gemma)

and layer 32 of 64 (Qwen) — canonical mid-stack layers from the `assistant_axis` library (Lu et al., 2026) — via `ActivationSteering` hook in *addition* mode at all token positions. Models loaded in `bfloat16` on H100 GPUs.

3.2. Steering mechanism and metric

For a unit-normalized steering vector v and scalar coefficient α , steered residual-stream activation at the target layer is

$$h'_\ell = h_\ell + \alpha v. \quad (1)$$

We measure the sycophancy logit

$$\text{syc_logit} = \log p(\text{syc_token}) - \log p(\text{hon_token}) \quad (2)$$

at the final prompt position, where `syc_token` matches the user’s stated opinion. Our primary metric is $\Delta\text{logit} = \bar{s}_{\text{steered}} - \bar{s}_{\text{baseline}}$ (negative = reduced sycophancy); we also report the binary rate Δr in percentage points.

3.3. Conditions

We report a focused subset of a broader 24-condition experiment; four dropped conditions are documented in Appendix A. The CAA baseline is extracted following Rimsky et al. (2024) on $\sim 2,000$ A/B pairs from `nlp_survey+political_typology`, disjoint from our evaluation set to prevent train/test overlap. Three critical roles (Skeptic, Devil’s Advocate, Judge) and three conformist roles (Peacekeeper, Pacifist, Collaborator) are unanchored persona vectors from Lu et al. (2026), computed as `unit(role - default)` and steered with positive coefficient *toward* the role. We use unanchored rather than anchored directions because we aim to shift the model away from its sycophantic default, not isolate role-specific distinctiveness. Ten random unit vectors sampled from an isotropic Gaussian and normalized, pooled as a null baseline.

3.4. Benchmark and splits

The evaluation benchmark is `philpapers2020` (Perez et al., 2022): 300 base questions $\times$ 2 orderings (counterbalancing Gemma’s 93% A-bias) = 600 rows per seed. We enforce a 50/50 tune/test split (seed 99, pairs kept together). Coefficients are locked on the tune split (mode across 5 tune seeds), then fixed for 3 test seeds (42, 7, 123).

3.5. Coefficient sweep

Gemma: $\{\pm 5000, \pm 2000, \pm 1000, \pm 500, 0\}$; and Qwen: $\{\pm 500, \pm 200, \pm 100, \pm 50, 0\}$. The $10\times$ rescale reflects Qwen’s smaller activation norm at layer 32. Degradation is flagged when the steered rate collapses to ≈ 0.5 and the logit approaches the random-mean band.

3.6. Statistical testing

Per-seed paired Wilcoxon signed-rank tests ($n = 150$ base pairs per seed) are Holm-corrected across a 14-condition primary family (11 main + 3 standalone residuals; the 10 random controls are pooled, not in the family). Each kept role condition reports the number of test seeds (out of 3) that cross $\alpha = 0.05$ after correction. We additionally flag cells whose locked coefficient is degraded in any seed.

4. Experiments

4.1. Critical Roles Reduce Sycophancy

Table 1 and Figure 6 present the primary results.

Gemma 2 27B (baseline logit +1.01, rate 59%). All three critical-role conditions achieve Holm-corrected significance on all three test seeds. The critical-family mean Δlogit is -0.596 , reaching 68% of CAA’s -0.879 . Skeptic achieves a 9.6-pp binary-rate reduction, slightly exceeding CAA’s 8.9 pp despite using no sycophancy-specific data. Devil’s Advocate ($\Delta\text{logit} = -0.521$, $\Delta r = -8.7$ pp) and Judge (-0.556 , -9.3 pp) show similarly robust effects. The random null (-0.254 , -2.1 pp) is substantially smaller, confirming that critical-role effects are direction-specific rather than artifacts of activation perturbation at comparable norm.

Qwen 3 32B (baseline logit +3.00, rate 84%). Absolute effect sizes are larger, consistent with the higher baseline. The critical-family mean Δlogit is -1.931 , reaching 98% of CAA’s -1.965 . Devil’s Advocate ($\Delta\text{logit} = -2.272$) *numerically exceeds* CAA. Skeptic (-1.823 , -18.1 pp) and Judge (-1.699 , -4.4 pp) are strongly significant on all seeds. The random null is larger on Qwen (-1.058 , -10.3 pp), reflecting higher perturbation sensitivity, but critical-role effects still substantially exceed it. Per-seed consistency is high: Skeptic std = 0.013 on Gemma and 0.058 on Qwen (Appendix B).

4.2. Conformist Roles: Heterogeneous Effects

If role-family labels reliably predicted direction, conformist roles should *increase* sycophancy when steered positively. Instead, effects are weak and heterogeneous.

On Gemma, the conformist-family mean Δlogit is $+0.031$ (range $[-0.052, +0.100]$), indistinguishable from noise. Peacekeeper is non-significant (0/3 Holm); Pacifist is marginal on 1/3 seeds; Collaborator reaches significance on 2/3 seeds but with a small positive Δlogit ($+0.045$). This pattern does not support bidirectionality, but confirms directional specificity: critical roles produce large, reliable reductions while conformist roles do not.

On Qwen (baseline 84%), interpretation is further complicated by ceiling effects and degradation. Pacifist at

Table 1. Results at tune-locked coefficients on the held-out test split (3 seeds). Δr in percentage points. Qwen Pacifist omitted (degraded at +500).

CONDITION	COEF	$\Delta\text{log} \pm \text{sd}$	Δr	SIG
<i>Gemma 2 27B</i>				
CAA (TARGETED)	-2k	$-0.879 \pm .001$	-8.9	—
SKEPTIC	+2k	$-0.711 \pm .013$	-9.6	—
DEVIL’S ADV.	+2k	$-0.521 \pm .016$	-8.7	—
JUDGE	+2k	$-0.556 \pm .003$	-9.3	—
PEACEKEEPER	+2k	$-0.052 \pm .004$	+1.8	NS
PACIFIST	+2k	$+0.100 \pm .001$	+0.3	—
COLLABORATOR	+500	$+0.045 \pm .006$	+1.7	—
RANDOM ($n = 10$)	—	$-0.254 \pm .006$	-2.1	—
<i>Qwen 3 32B</i>				
CAA (TARGETED)	-200	$-1.97 \pm .126$	-20.9	—
SKEPTIC	+200	$-1.82 \pm .058$	-18.1	—
DEVIL’S ADV.	+200	$-2.27 \pm .195$	-16.6	—
JUDGE	+200	$-1.70 \pm .075$	-4.4	—
PEACEKEEPER	-200	$-0.709 \pm .108$	+0.1	NS
COLLABORATOR	-100	$-0.029 \pm .016$	-1.7	NS
RANDOM ($n = 10$)	—	$-1.058 \pm .077$	-10.3	—

Gemma Cosine Similarity Between Kept Steering Vectors

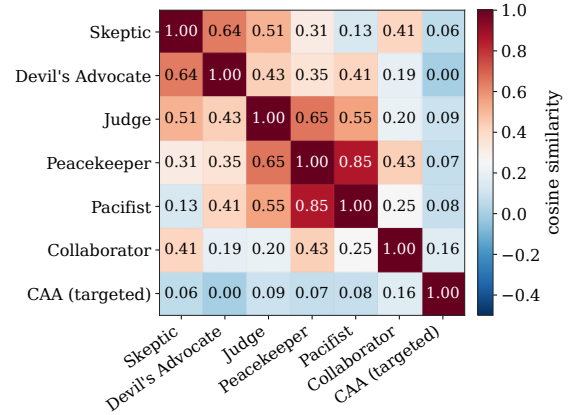


Figure 1. Cosine similarity heatmap. Critical roles cluster ($\cos \approx 0.6$ – 0.7); conformist roles cluster separately ($\cos \approx 0.8$). All role–CAA cosines < 0.17 , but *signs* differ across models.

+500 produces model collapse (repetitive loops: “*the truth that is the truth. . .*”) and is flagged as degraded. Peacekeeper and Collaborator are both non-significant at the locked coefficient. Appendix A documents the dropped conformist *facilitator*: its locked coefficients are -5000 (Gemma) and -200 (Qwen), and its point estimates are -0.727 and -0.469 — but *neither is Holm-significant at any seed* ($p_{\text{adj}} = 1.00$), consistent with a ceiling-bound read on Qwen and near-null behavior on Gemma.

4.3. Geometric Relationship to CAA

Figure 5 shows cosine similarities between steering vectors. All role–CAA cosines fall below 0.17 on Gemma and below 0.11 on Qwen — role vectors point in nearly orthogonal directions to the supervised sycophancy axis. Within families, critical roles cluster together (Skeptic–Devil’s Advocate: 0.64 Gemma, 0.71 Qwen) and conformist roles cluster sepa-